Supplement to: A Frame-Only Census of Game Mechanics Built on ARC-AGI-3 Failed Preregistered Transfer Tests on MiniGrid/BabyAI and Procgen

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This supplement accompanies the article of the same title. The article cites it as "Supplement, Section S.n". It holds the full registered tables, the amendment trail, the exclusion lists, the provenance of every number and the hashes of the frozen files, the exploratory ARC-AGI-3 line on the 17 held-out games, and (Section S.8) the passages of the method and the exploratory diagnosis that the article gives in condensed form. Everything here is copied from committed files of the repository, and every number in it is checked against the same list of values as the article. Repository addresses, commit hashes, registration addresses and preprint identifiers are withheld in this version for double-anonymous review.

S.1 The Full V1 Table

Table S1 is docs/tables/v1.md, as written by the frozen analysis code at prereg-v1.1, with every row and cell unchanged. Agreement is the share of games where the census verdict equals the truth, and kappa is given only where both truth values have at least 3 games. The code's identifiers are kept. On MiniGrid/BabyAI, "macro" is absolute moves and "raw" primitive actions. On Procgen, "center_off" is the whole-level view and "center_on" the view centered on the agent, "in_range" is the nine games of the operating range, and "camera_distinct" the five games where the two views differ.

TABLE S1
Test V1, every line, verbatim from docs/tables/v1.md.

suite	condition	subset	assumption	n	truth_holds	agreement	kappa	verdict
minigrid	macro	all	A1	161	161	0.9876		PASS
minigrid	macro	all	A2	161	161	0.9876		PASS
minigrid	macro	all	A3	161	161	0.4286		FAIL
minigrid	macro	all	A4	161	156	0.3354	-0.0435	FAIL
minigrid	macro	all	A5	161	36	0.2671	-0.4125	FAIL
minigrid	raw	all	A1	161	161	0.9379		PASS
minigrid	raw	all	A2	161	161	0.4596		FAIL
minigrid	raw	all	A3	161	0	0.5466		FAIL
minigrid	raw	all	A4	161	156	0.0621	0.0021	FAIL
minigrid	raw	all	A5	161	36	0.0994	-0.3374	FAIL
procgen	center_off	all	A1	14	14	0.9286		PASS
procgen	center_off	all	A2	14	6	0.5714	0.2222	FAIL
procgen	center_off	all	A3	14	2	0.5714		FAIL
procgen	center_off	all	A4	14	13	0.0714		FAIL
procgen	center_off	all	A5	14	6	0.0714	-0.7843	FAIL
procgen	center_off	in_range	A1	9	9	0.8889		neither
procgen	center_off	in_range	A2	9	4	0.4444	0.0000	FAIL
procgen	center_off	in_range	A3	9	2	0.4444		FAIL
procgen	center_off	in_range	A4	9	8	0.1111		FAIL
procgen	center_off	in_range	A5	9	2	0.1111		FAIL
procgen	center_on	camera_distinct	A1	5	5	0.8000		reported, not tested
procgen	center_on	camera_distinct	A2	5	2	0.8000		reported, not tested
procgen	center_on	camera_distinct	A3	5	0	0.6000		reported, not tested
procgen	center_on	camera_distinct	A4	5	5	0.0000		reported, not tested
procgen	center_on	camera_distinct	A5	5	4	0.0000		reported, not tested

S.2 The Full V2 Tables

Table S2 is docs/tables/v2.md, as written by the frozen analysis code at prereg-v1.1. Its thirteen columns do not fit the page, so it is printed in two parts with the same rows in the same order: Table S2a gives each line and its verdict, Table S2b its

statistics. No row, cell or value is changed. The agents are R, the explorer, N, the planner (instrument check), and I1, the graph-based explorer. Lines marked "primary" exclude the environments of Section S.4, "primary, D9b included" puts back the exclusions specific to I1, and "all games" includes every environment. auroc_B is the AUROC of the nuisance baseline, ci_low and ci_high the 95% bootstrap interval, and balanced_acc the balanced accuracy of the conjunctive prediction. Condition identifiers are as in Table S1; "native" is ARC-AGI-3. The two ARC-AGI-3 lines are computed over all 25 public games because of the defect described in the article (Section III-J), and the "FAIL" printed on them has no registered meaning.

TABLE S2a
TEST V2, THE LINES AND THEIR VERDICTS, FROM DOCS/TABLES/V2.MD.

suite	condition	subset	agent	role	n	positives	verdict
arc	native	all	N	reported	25	14	FAIL
arc	native	all	R	reported	25	15	FAIL
minigrid	macro	primary	I1	evidence	83	61	neither
minigrid	macro	primary	N	instrument check	112	100	neither
minigrid	macro	primary	R	evidence	112	101	VOID
minigrid	macro	primary, D9b included	I1	evidence	112	61	reported, not tested
minigrid	macro	all games	I1	evidence	161	61	reported, not tested
minigrid	macro	all games	N	instrument check	161	100	reported, not tested
minigrid	macro	all games	R	evidence	161	101	reported, not tested
minigrid	raw	primary	I1	evidence	98	71	FAIL
minigrid	raw	primary	N	instrument check	155	130	FAIL
minigrid	raw	primary	R	evidence	155	126	FAIL
minigrid	raw	primary, D9b included	I1	evidence	155	71	reported, not tested
minigrid	raw	all games	I1	evidence	161	71	reported, not tested
minigrid	raw	all games	N	instrument check	161	130	reported, not tested
minigrid	raw	all games	R	evidence	161	126	reported, not tested
procgen	center_off	all	I1	evidence	14	6	FAIL
procgen	center_off	all	N	instrument check	14	9	neither
procgen	center_off	all	R	evidence	14	9	neither
procgen	center_off	in_range	I1	evidence	9	4	neither
procgen	center_off	in_range	N	instrument check	9	5	neither
procgen	center_off	in_range	R	evidence	9	5	neither
procgen	center_on	camera_distinct	I1	evidence	5	3	reported, not tested
procgen	center_on	camera_distinct	N	instrument check	5	4	reported, not tested
procgen	center_on	camera_distinct	R	evidence	5	4	reported, not tested

The analysis code also writes one verdict line per suite, given in Table S3 verbatim from docs/tables/v2.json.

S.3 THE AMENDMENT TRAIL, D1 TO D16

Each entry is condensed from the diff sections of the registered protocol, where the full text, the file and line references and the dates are given. D1 to D15 were written on 27 September 2026 and registered at prereg-v1.1; D16 was written on 13 October 2026 and registered at prereg-v1.2.

D1, six actions at most. The frozen explorer presses key identifiers 1, 2, 3, 4, 5 and 7 only, so each suite is presented through a fixed key map: turn left, turn right, forward, pickup, drop and toggle under primitive MiniGrid/BabyAI actions; north, south, west, east, pickup and toggle under absolute moves; and up, down, left, right and two special actions on Procgen. An environment with fewer actions keeps only those it has.

D2, canvas placement. A MiniGrid/BabyAI grid sits one row down at the top left of a 64×64 canvas, and every other cell holds a reserved padding index, because the frozen code reads rows 1 to 62 only and anchors the frame at the top left.

D3, the Procgen border rows. The 64×64 Procgen frame fills the canvas, so the frozen code does not see its top and bottom pixel rows. This is recorded and not corrected.

D4, what a reset does. On MiniGrid/BabyAI a reset replays the current layout, and the layout advances only on a success. Procgen cannot restart a level, so a reset there takes no step and returns the current frame, and it still counts as an action.

D5, the wall-clock cap. The cap per suite is three times the longest pilot run, rounded up to a whole minute and computed by a script: 3300 s for MiniGrid/BabyAI and 7500 s for Procgen. Pilot runs were uncapped. A run cut by the cap is kept and reported as cut; none was cut.

D6, the runner. One registered script runs both suites, calls the author's agents exactly as the ARC-AGI-3 runner does, refuses to run if any frozen file differs from its registered hash, and writes one row per run.

D7, the boundary of I1. I1 runs unchanged behind an adapter that offers it key identifiers 1 to 5 only, shifts palette values that collide with its status-bar marker, and caps the level count it sees. It therefore lacks toggle on MiniGrid/BabyAI and the second special action on Procgen.

TABLE S2b
TEST V2, THE STATISTICS OF THE SAME LINES, IN THE SAME ORDER, FROM DOCS/TABLES/V2.MD.

suite	condition	subset	agent	auroc	ci_low	ci_high	balanced_acc	auroc_B
arc	native	all	N	0.4188	0.1978	0.6379	0.5357	0.5000
arc	native	all	R	0.4100	0.1850	0.6354	0.5333	0.5000
minigrid	macro	primary	I1	0.7187	0.5902	0.8298	0.5838	0.8413
minigrid	macro	primary	N	0.6613	0.5049	0.7941	0.5700	0.8725
minigrid	macro	primary	R	0.6985			0.5693	0.8515
minigrid	macro	primary, D9b included	I1	0.7719	0.6840	0.8529	0.5968	0.8734
minigrid	macro	all games	I1	0.7517	0.6764	0.8223	0.6016	0.7562
minigrid	macro	all games	N	0.6352	0.5514	0.7114	0.5700	0.5752
minigrid	macro	all games	R	0.6434	0.5609	0.7208	0.5693	0.5589
minigrid	raw	primary	I1	0.5370	0.4324	0.6410	0.5070	0.8534
minigrid	raw	primary	N	0.5255	0.4320	0.6156	0.5038	0.6202
minigrid	raw	primary	R	0.5224	0.4330	0.6096	0.5040	0.6561
minigrid	raw	primary, D9b included	I1	0.6258	0.5405	0.7065	0.5070	0.8184
minigrid	raw	all games	I1	0.6069	0.5235	0.6926	0.5070	0.8086
minigrid	raw	all games	N	0.4772	0.3829	0.5696	0.5038	0.6036
minigrid	raw	all games	R	0.4805	0.3919	0.5687	0.5040	0.6472
procgen	center_off	all	I1	0.6458	0.3333	0.9445	0.5000	0.5104
procgen	center_off	all	N	0.7222	0.5000	0.9333	0.5000	0.5667
procgen	center_off	all	R	0.7222	0.5000	0.9333	0.5000	0.5667
procgen	center_off	in_range	I1	0.8750	0.5000	1.0000	0.5000	0.4750
procgen	center_off	in_range	N	0.8000	0.5000	1.0000	0.5000	0.7000
procgen	center_off	in_range	R	0.8000	0.5000	1.0000	0.5000	0.7000
procgen	center_on	camera_distinct	I1	0.3333	0.0000	1.0000	0.5000	0.6667
procgen	center_on	camera_distinct	N	0.3750	0.0000	0.8333	0.5000	1.0000
procgen	center_on	camera_distinct	R	0.3750	0.0000	0.8333	0.5000	1.0000

TABLE S3
THE V2 SUITE LINES, VERBATIM FROM DOCS/TABLES/V2.JSON.

suite	condition	subset	suite line
arc	native	all	reported, not tested
minigrid	macro	primary	not failed (decided by I1 alone)
minigrid	macro	primary, D9b included	reported, not tested
minigrid	macro	all games	reported, not tested
minigrid	raw	primary	suite FAIL (decided by R and I1)
minigrid	raw	primary, D9b included	reported, not tested
minigrid	raw	all games	reported, not tested
procgen	center_off	all	not failed (decided by R and I1)
procgen	center_off	in_range	not failed (decided by R and I1)
procgen	center_on	camera_distinct	reported, not tested

D8, infrastructure checks. The fidelity check, the pilot check, the Procgen camera check and the determinism check play random actions or read existing rows. None computes a census, an agent decision or a test statistic.

D9, exclusions for R and N. An environment is excluded from an agent's primary V2 line when, on every seed, success cannot be reached without an action missing from the agent's key map, either as the success action or as a precondition of every successful episode. The environment with no success at all is on every list. The lists hold 6 environments under primitive actions and 49 under absolute moves.

D9b, exclusions for I1. After the pilot showed I1 completing no level on a locked-door environment, the D9 rule was applied to I1's own key map, adding the 57 environments where success needs toggle on every seed. I1's lists hold 63 and 78 environments. Procgen has no exclusion, because no level end needs the second special action.

D10, seeding text. The v1 seeding text for MiniGrid/BabyAI is replaced by D4, and the runner's commands are added.

D11, the analysis code. The blob hashes of the analysis scripts at the tag are recorded; the scripts changed only to implement D9, D9b and D13. The synthetic tests passed, 20 of 20.

D12, the ARC-AGI-3 cap. The registered ARC-AGI-3 baseline had inherited an unregistered clock of 900 s per run, which cut eight runs short. The cap is 3600 s per run, registered at prereg-v1.1. The eight runs were repeated under the v1 tag, and the repeats replaced the cut rows.

D13, camera identity. Ten Procgen games set the camera option themselves at every level start, so the camera-on condition draws the same frames as the camera-off condition on the nine confirmatory games among them. The camera-on condition is reported on the five other games only and never tested; on the nine, the camera-on runs must equal the camera-off runs, which

is checked as a matter of determinism.

D14, no adapter change after the pilot. The pilot showed no errors, no cut runs and the expected identity between conditions, and no adapter was changed.

D15, G by a registered replay. The rows for the goal-identification cost G were to come from a separate replay of N, registered as a further version before any replay, with every replay required to reproduce the confirmatory actions exactly.

D16, G is not measured. The replay of D15 is withdrawn after the outcome, because the instrument that G would accompany failed V1 on both suites. G remains defined in the protocol, and no value of it is reported.

S.4 The Exclusion Lists

These lists are the registered constants in scripts/analyze_common.py at prereg-v1.1. The source reason for each identifier is given in the protocol under D9 and D9b.

B1. No success at all (1 environment; on every agent's list in both conditions):

MiniGrid-Playground-v0.

B2. Success needs "done" (5; on every list in both conditions; "done" is in no key map):

MiniGrid-GoToDoor-5x5-v0; MiniGrid-GoToDoor-6x6-v0; MiniGrid-GoToDoor-8x8-v0; MiniGrid-GoToObject-6x6-N2-v0; MiniGrid-GoToObject-8x8-N2-v0.

B3. Success is a drop (15; on the absolute-move lists of every agent):

MiniGrid-PutNear-6x6-N2-v0; MiniGrid-PutNear-8x8-N3-v0; BabyAI-PutNextLocal-v0; BabyAI-PutNextLocalS5N3-v0; BabyAI-PutNextLocalS6N4-v0; BabyAI-PutNextS4N1-v0; BabyAI-PutNextS5N1-v0; BabyAI-PutNextS5N2-v0; BabyAI-PutNextS5N2Carrying-v0; BabyAI-PutNextS6N3-v0; BabyAI-PutNextS6N3Carrying-v0; BabyAI-PutNextS7N4-v0; BabyAI-PutNextS7N4Carrying-v0; BabyAI-MoveTwoAcrossS5N2-v0; BabyAI-MoveTwoAcrossS8N9-v0.

B4. A drop is a precondition (28; on the absolute-move lists of every agent, and on I1's lists in both conditions as toggle preconditions):

MiniGrid-KeyCorridorS3R1-v0; MiniGrid-KeyCorridorS3R2-v0; MiniGrid-KeyCorridorS3R3-v0; MiniGrid-KeyCorridorS4R3-v0; MiniGrid-KeyCorridorS5R3-v0; MiniGrid-KeyCorridorS6R3-v0; MiniGrid-UnlockPickup-v0; MiniGrid-BlockedUnlockPickup-v0; MiniGrid-ObstructedMaze-1Dl-v0; MiniGrid-ObstructedMaze-1Dlh-v0; MiniGrid-ObstructedMaze-1Dlhb-v0; MiniGrid-ObstructedMaze-1Q-v1; MiniGrid-ObstructedMaze-2Dl-v0; MiniGrid-ObstructedMaze-2Dlh-v0; MiniGrid-ObstructedMaze-2Dlhb-v1; MiniGrid-ObstructedMaze-2Q-v1; MiniGrid-ObstructedMaze-Full-v1; BabyAI-KeyCorridor-v0; BabyAI-KeyCorridorS3R1-v0; BabyAI-KeyCorridorS3R2-v0; BabyAI-KeyCorridorS3R3-v0; BabyAI-KeyCorridorS4R3-v0; BabyAI-KeyCorridorS5R3-v0; BabyAI-KeyCorridorS6R3-v0; BabyAI-UnlockPickup-v0; BabyAI-UnlockPickupDist-v0; BabyAI-BlockedUnlockPickup-v0; BabyAI-UnlockToUnlock-v0.

B5. Success is a toggle (20; on I1's lists in both conditions):

MiniGrid-Unlock-v0; MiniGrid-RedBlueDoors-6x6-v0; MiniGrid-RedBlueDoors-8x8-v0; BabyAI-Open-v0; BabyAI-OpenDoor-v0; BabyAI-OpenDoorColor-v0; BabyAI-OpenDoorDebug-v0; BabyAI-OpenDoorLoc-v0; BabyAI-OpenDoorsOrderN2-v0; BabyAI-OpenDoorsOrderN2Debug-v0; BabyAI-OpenDoorsOrderN4-v0; BabyAI-OpenDoorsOrderN4Debug-v0; BabyAI-OpenRedBlueDoors-v0; BabyAI-OpenRedBlueDoorsDebug-v0; BabyAI-OpenRedDoor-v0; BabyAI-OpenTwoDoors-v0; BabyAI-Unlock-v0; BabyAI-UnlockLocal-v0; BabyAI-UnlockLocalDist-v0; BabyAI-KeyInBox-v0.

B6. A toggle is a precondition (these 9, plus the 28 of B4; on I1's lists in both conditions):

MiniGrid-DoorKey-5x5-v0; MiniGrid-DoorKey-6x6-v0; MiniGrid-DoorKey-16x16-v0; MiniGrid-LockedRoom-v0; MiniGrid-MultiRoom-N2-S4-v0; MiniGrid-MultiRoom-N4-S5-v1; MiniGrid-MultiRoom-N6-v0; BabyAI-GoToImpUnlock-v0; BabyAI-PickupAbove-v0.

TABLE S4
THE EXCLUSION LISTS PER AGENT AND CONDITION.

agent and condition	lists	excluded	primary games
R and N, primitive actions	B1, B2	6	155
R and N, absolute moves	B1, B2, B3, B4	49	112
I1, primitive actions	B1, B2, B4, B5, B6	63	98
I1, absolute moves	B1, B2, B3, B4, B5, B6	78	83
any agent, Procgen	none	0	14

S.5 Provenance of the Numbers

Every number in the article and in this supplement is a value in docs/manuscript_tog/numbers.json, which scripts/manuscript_numbers.py writes from committed files, by the build described in docs/manuscript_tog/BUILD.md. scripts/manuscript_check.py, run with the options –dir docs/manuscript_tog –ieee, rejects any numeral in either file that is

not one of those values. The only exceptions are the fields of the reference entries, which were verified against the publisher's or arXiv's page and are recorded, entry by entry, in docs/manuscript_tog/REFERENCES_CHECK_tog.md; the check accepts such an entry only if it equals the recorded entry character for character. Table S5 lists, for each class of number, the committed files its values come from. It is written by scripts/manuscript_numbers.py and copied here unchanged. In this version the commit that last changed each file is withheld, as every commit hash is, for double-anonymous review.

TABLE S5
NUMBER CLASSES AND THEIR SOURCES.

number class	source files
count	docs/EXPLORATORY_V1_DIAGNOSIS.md; docs/GROUND_TRUTH_minigrid.md; docs/GROUND_TRUTH_procgen.md; docs/HELDOUT.md; docs/PREREG.md; docs/STATE.md; docs/tables/v1.json; docs/tables/v2.json; runs/checks/determinism_check.json; runs/exploratory/v1_diagnosis.json; runs/pilot_checks/pilot_check.json; runs/pilot_checks/procgen_camera_check.json; scripts/analyze_common.py; reference record of the second earlier version (ARC-AGI-3 report and added references)
rate (agreement, kappa, AUROC, interval bound, share)	docs/tables/v1.json; docs/tables/v2.json; runs/exploratory/v1_diagnosis.json
registered threshold	docs/PREREG.md
constant, cap, seed or time	docs/PREREG.md; docs/tables/v2.json; runs/pilot_checks/pilot_check.json; runs/pilot_checks/procgen_camera_check.json; reference record of the second earlier version (ARC-AGI-3 report and added references)
commit or blob hash	docs/PREREG.md; docs/STATE.md; git tag objects and git ls-tree
date	docs/HELDOUT.md; docs/PREREG.md; docs/STATE.md; git tag objects and git ls-tree; reference record of the second earlier version (ARC-AGI-3 report and added references)
package version	docs/PREREG.md
URL or DOI	docs/STATE.md; docs/preprint/preprint.md; docs/preprint/references.bib; reference record of the first earlier version of this article; reference record of the second earlier version (ARC-AGI-3 report and added references); the author's review of the first earlier version (reference URLs)
reference field, verdict word or ORCID	docs/STATE.md; docs/preprint/preprint.md; docs/preprint/references.bib; docs/tables/v1.json; docs/tables/v2.json; reference record of the first earlier version of this article; reference record of the second earlier version (ARC-AGI-3 report and added references)

The sources of the article's figures, tables and constants are these. Figures 1 and 2 are drawn by scripts/manuscript_figures.py from docs/tables/v1.json and docs/tables/v2.json; Figures 3 and 4 by the same script from runs/exploratory/v1_diagnosis.json, written by scripts/explore_v1_diagnosis.py, which calls the frozen analysis functions unchanged. Tables I and II of the article are the verdict columns of Tables S1 and S2a; Table III is the outcome table of the registered protocol. The census constants are registered with their file, line and commit: LEARN_AGREE = 3 at scripts/g_nav.py:55, the 2:1 margin at scripts/g_nav.py:756, LEARN_MAX = 8 at scripts/g_nav.py:56 and the search limit of 40000 nodes at scripts/g_nav.py:54, and the motion threshold of 2 cells at scripts/g_motion.py:20. The ARC-AGI-3 split, with the move of dc22, is in docs/HELDOUT.md. The registered texts of the two OSF registrations are in the repository as docs/OSF_REGISTRATION_final.txt and docs/OSF_REGISTRATION_v1_1_final.txt. The agent I1 is in the directory external/rudakov_just_explore, which is not covered by the repository's license and keeps the upstream MIT notice of its own authors.

S.6 FROZEN FILES AND ANALYSIS SCRIPTS

The six census files were frozen at prereg-v1 and are identical at prereg-v1.1; the runner checked their blob hashes before every run. The analysis scripts are those at prereg-v1.1. Each hash below equals the output of git ls-tree at prereg-v1.1.

S.7 THE EXPLORATORY ARC-AGI-3 LINE ON THE 17 HELD-OUT GAMES

This section is exploratory. Table S7 gives the V2 line on the 17 held-out ARC-AGI-3 games, computed after the outcome by the same frozen functions, from runs/exploratory/v1_diagnosis.json. It is reported and never tested. ARC-AGI-3 rows carry no adapter fields, so the baseline's features are constant there and its AUROC is 0.5000 by construction.

S.8 PASSAGES CONDENSED IN THE ARTICLE

The article gives the following passages in shorter form, to meet the journal's length limit. They are reproduced here as they stood in the earlier version of the article, and they add no result. The paragraphs on the exploratory diagnosis are exploratory, as in the article.

TABLE S6
GIT BLOB HASHES OF THE FROZEN FILES AND THE ANALYSIS SCRIPTS.

file	role	git blob hash
scripts/g_agent.py	explorer R, frozen	72585a4dbb926d4512ff9596f58118a176bb0646
scripts/g_nav.py	planner N and census report, frozen	a8a7eb61e62a658c239b98e0224e3588a662d1a0
scripts/g_motion.py	motion detector, frozen	14018512a5c6b0efc81aed61c5466654d338be0b
scripts/g_control_probe.py	key set, frozen	d5ff2a4dc77b9e1148929e7db03fe99a9d2b0bf8
scripts/g_nav_run.py	ARC-AGI-3 runner, frozen	2cc0146a697e525e5873d4e29cb80f4ae8bd6297
scripts/g_census_table.py	ARC-AGI-3 census table rule, frozen	488feaca9c10985c4d0ff62791ceb4fa6fa6b9a8
scripts/analyze_common.py	census rules, statistics, exclusion lists	caad22731807db78aff9c92c993a79a63f042ec3
scripts/analyze_v1.py	test V1	515f44b0f007182172022272383de548e80d5a11
scripts/analyze_v2.py	test V2	90a849c44e47addd8efdbfa579efdb16288edd50
scripts/analyze_g.py	cost G (not run)	46bf991d86400f0189d8898a7e068044d25c629a
scripts/determinism_check.py	determinism check	4f98f6430e9221deb1a43f2313eb00f5ca676969

TABLE S7
EXPLORATORY V2 LINE ON THE 17 HELD-OUT ARC-AGI-3 GAMES. REPORTED, NEVER TESTED.

agent	n	positives	AUROC	95% interval	balanced accuracy	baseline AUROC
N	17	8	0.3542	[0.1041, 0.6346]	0.5000	0.5000
R	17	9	0.3403	[0.0909, 0.6213]	0.5000	0.5000

Presentation of the benchmarks. On MiniGrid/BabyAI, “drop” is not offered under absolute moves. On Procgen, frames are mapped to integer indices by exact color and capped at 64 indices. The camera check found 401 of 401 frames identical between the two conditions on the ten games that set the camera option themselves, and a difference from the first frame on the other six.

The amendment trail, in the order of the diff. D1 to D4 fix how each benchmark is presented through the six-key limit, D5 sets the per-run time caps, D6 the runner, D7 the boundary of I1 and D8 the infrastructure checks. D9 and D9b set the exclusions from V2, and D10 and D11 the analysis code at the tag. D12 sets a time cap for ARC-AGI-3 and the repeat of eight ARC runs that an unregistered clock had cut short. D13 records the Procgen camera identity, D14 the absence of any adapter change after the pilot, and D15 a separate replay planned for G (Section S.3).

The pilot check. It read the 480 pilot runs, 360 on MiniGrid/BabyAI and 120 on Procgen, and found 0 errors and 0 runs short of the budget. It also projected the running time of every confirmatory configuration and found 0 over its cap, the largest projections being 1517.8 s and 2550.0 s.

Exploratory consistency check. The 25-game ARC-AGI-3 lines recomputed by the same functions equal those of the registered table to four decimals.

Exploratory, A1. The census agrees with the truth on 4 of 5 camera-on games. On the games where the truth holds and the census fails, most seeds failed the per-seed A1 rule, which fails when no key has a learned shift or when every tried key has the same shift. A1, input coupling, is the assumption that the census and the truth table describe in nearly the same terms, and it is the only one with no failing V1 line.

Exploratory, A2 and A3. On the 870 seeds of the 87 primitive-action games where the truth holds A2 and the census fails, forward was accepted on 208, so under primitive actions the census’s A2 rested on forward alone. On the 6 Procgen games where the census says A2 holds and the truth fails, it found one controlled object on at least 9 of 10 seeds. Of the 775 seeds of the 92 absolute-move games whose A3 verdict is “fails”, 755 carry the planner’s first-level outcome “mispredicted” and the other 20 never located the agent. Under primitive actions, one step vector per key, learned from the starting facing, cannot reveal a dependence on facing while the turns record no shift.

Exploratory, A4. On Procgen the counter of walked-through “walls” is above zero on some seeds of all 13 games where the truth holds A4, and on all 10 seeds of 5 of them. That counter rises when the planner walks through a color it had classified as a wall. The counters do not say whether that came from the game’s obstacles, from momentum or from the planner’s classification.

Exploratory, A5. The census said “holds” where the truth fails on 0 Procgen games. The MiniGrid/BabyAI environments where it did so belong to the GoTo, GoToObj, OpenDoor, OneRoom, Fetch and Unlock families.

Exploratory, seed dissent. Dissent is rare for A4 on Procgen, on 2 of 14 games.

The nuisance baseline, in the discussion of the earlier version. Three of the six primary MiniGrid/BabyAI lines on which the baseline ranked the games better than the census are tested by the registered text and three are reported. On Procgen the baseline did not rank the games better than the census on any line of the whole-level view.